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Declaration:

I affirm that I have identified all my sources and that no part of my dissertation uses unacknowledged materials.

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Abstract

Ever since Harry Markowitz fundamentally reshaped the financial world in 1952, his Mean-Variance Framework has been treated as the "holy grail" of quantitative finance. It provided a elegant mathematical solution to a messy human problem: how much risk is "worth it" for a specific return? However, when we take this mid-century Western theory and drop it into the high-octane, often unpredictable environment of an emerging economy like India, we encounter what I call the **"Optimization Enigma."**

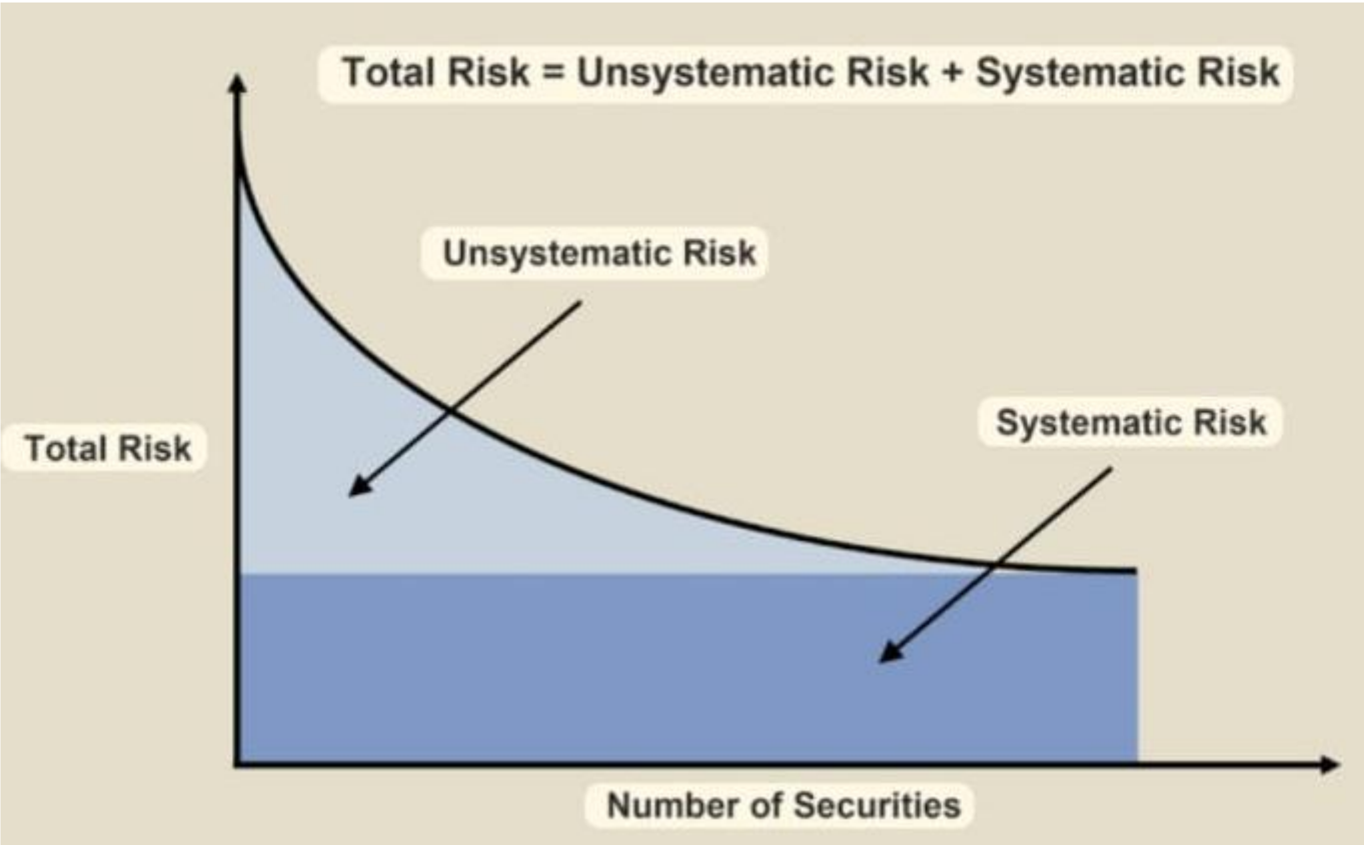
Most conventional investment models play it safe; they rely on monthly or quarterly data snapshots. While this makes the math look "cleaner" by smoothing out the jagged edges of market movement, it also hides the truth. In the Indian banking sector, the real story isn't told in months—it's told in minutes. By ignoring the daily "noise" and those sudden clusters of volatility that characterize the Nifty Bank index, traditional strategies effectively put on blinders.

They miss the stochastic "heartbeat" of the market—the rapid-fire reaction to a sudden RBI policy shift or a global liquidity crunch. This research argues that to truly understand risk in a market as vibrant and reactive as India’s, we cannot afford to "smooth over" the data. We have to dive into the daily chaos of the charts to find the real patterns hidden beneath the surface.

Introduction

The real reason investors need a framework like Modern Portfolio Theory (MPT) isn't just to "make money," but to solve the paralyzing dilemma of uncertainty. In the open market, every investor is haunted by two distinct types of ghosts: systematic and unsystematic risk. Unsystematic risk is the "specific" danger—the chance that a single company like SBI or Axis Bank suffers a management scandal or a bad quarterly report. This is a risk you can actually outsmart through diversification. By spreading your capital across different banks or sectors, you ensure that one company's disaster doesn't sink your entire ship. However, systematic risk is the "market" danger—things like global inflation, a sudden hike in RBI interest rates, or a geopolitical crisis. These are the waves that move the entire ocean; no matter how many different boats you own, they all rise or fall together. This is where MPT steps in. Instead of just telling you to "buy a lot of stocks," it provides a mathematical blueprint to balance these two forces. It teaches us that a portfolio is more than just a collection of assets; it’s a living system where the goal is to find the perfect "sweet spot"—the Efficient Frontier—where you are taking on the absolute minimum amount of total risk necessary to achieve the return you're aiming for. It effectively turns the chaos of the stock market into a calculated, manageable strategy.

MPT calculates the **Expected Return** of a portfolio by looking at the weighted average of every individual asset. But here is the catch: while the return is a simple average, the **Risk** is not. Because these banks don't move in perfect lockstep—perhaps SBI reacts differently to a government policy than HDFC does to a global interest rate shift—their prices don't always peak and trough at the exact same moment. MPT exploits these tiny gaps in timing, known as **Covariance**, to cancel out a portion of the volatility. It's the financial equivalent of noise-cancelling headphones; by playing two opposing "waves" of risk against each other, the theory creates a smoother, quieter ride for the investor.
Building on this idea, Modern Portfolio Theory emphasizes the importance of diversification as the key mechanism for reducing unsystematic risk. By combining assets that do not perfectly correlate with one another, an investor can construct a portfolio that achieves a more favorable risk-return trade-off than any single asset alone. This leads to the concept of the *efficient frontier*, which represents the set of optimal portfolios that offer the highest expected return for a given level of risk. Portfolios that lie below the efficient frontier are considered suboptimal, as they do not maximize returns relative to their risk. Conversely, portfolios on the frontier are deemed efficient because they are structured in a way that no additional return can be achieved without increasing risk. Investors, depending on their individual risk tolerance, can choose a point along this frontier that aligns with their financial goals.



The above figure shows the dynamics between systematic and unsystematic risk where total risk is the combination of systematic and unsystematic risk. Systematic risk is the risk any investor is bound to take irrespective of his/her portfolio design and unsystematic risk is the amount of risk that can be minimized by diversifying his/her portfolio of asset

Data description

Ticker	AXISBANK.NS	HDFCBANK.NS	ICICIBANK.NS	SBIN.NS
Date				
2024-01-02	-0.004062	0.000589	-0.017309	-0.002967
2024-01-03	0.004882	-0.015540	0.001831	0.006236
2024-01-04	0.022323	0.010673	0.003145	-0.001089
2024-01-05	0.011989	-0.005129	0.006613	-0.001245
2024-01-08	-0.013236	-0.011209	-0.011844	-0.023564
...	...	...	...	...
2026-02-23	0.013358	0.012804	0.003508	0.009575
2026-02-24	0.000649	-0.014285	-0.010488	-0.003672
2026-02-25	0.011037	-0.003190	0.011274	-0.019147
2026-02-26	-0.005360	-0.009966	0.003137	0.007802
2026-02-27	-0.008347	-0.012148	-0.018680	-0.006470

535 rows × 4 columns

This dataset serves as the empirical heartbeat of my research, capturing the day-to-day market sentiment for the primary pillars of the Indian banking sector: Axis Bank, HDFC Bank, ICICI Bank, and SBI. By transforming raw price fluctuations into logarithmic returns of closing prices over these 535 trading days, I've essentially "cleaned" the data, stripping away the mathematical distortions of traditional percentage changes to ensure the continuous compounding required for rigorous portfolio modeling. Looking at the spread from January 2024 through February 2026, the numbers reveal a fascinating story of sectoral interdependence; you can see clear "volatility clusters" where global macro-shocks dragged all four giants into the red, contrasted against quieter days where an idiosyncratic move in one bank helped buffer losses in another. Ultimately, this matrix isn't just a table of decimals—it is the raw, high-frequency "noise" of the Nifty Bank index that the study now interrogates to find the mathematical order of the Efficient Frontier.

Methodology

To understand the mechanics of the research paper, we need to pull back the curtain on the Python implementation. The algorithm doesn't just "guess" the best portfolio; it uses a computational brute-force method known as a **Monte Carlo Simulation** to map out every mathematical possibility for the chosen banking stocks.

Here is the step-by-step methodology interpreted from the technical logic of the provided code.

Phase 1: Data Normalization (The Foundation)

The algorithm begins by pulling raw daily closing prices for **Axis, HDFC, ICICI, and SBI** using the `yfinance` library. However, raw prices are difficult to compare. To fix this, the code applies a **Logarithmic Transformation** (`np.log(1 + pct_change)`).

Why this matters: In the world of high-frequency daily data, standard percentage changes can be misleading because they aren't "additive" (a 10% gain followed by a 10% loss doesn't bring you back to zero). Log returns solve this, creating a statistically "stationary" dataset that allows the math of MPT to function accurately.

Phase 2: Defining the Financial "Engine"

The code establishes two core functions that act as the mathematical engine for the entire study:

- **The Return Calculator (`portfolioreturn`):** This function takes a set of weights and calculates the weighted average of the mean daily returns. It then multiplies the result by **252** (the number of trading days in a year) to annualize the return, moving the perspective from "daily noise" to "yearly expectation".
- **The Risk Calculator (`portfoliostd`):** This is the most critical part of the code. It doesn't just look at how much a stock fluctuates; it uses the **Covariance Matrix** (`stocks_lr.cov()`). By performing a dot product of the weights and the covariance matrix, the algorithm captures how these four banks move *together*. This calculation is what allows the code to find "diversification benefits".

Phase 3: The Monte Carlo Simulation (Stochastic Modeling)

Instead of trying to solve a complex equation, the algorithm uses a "trial and error" approach on a massive scale.

1. The `weightscreator` function generates an array of four random numbers.

2. These numbers are normalized so that they always sum to **1.0 (100%)**.
3. The code then runs this loop **100,000 times**.

For every single iteration, it calculates the specific risk and return for that random combination of Axis, HDFC, ICICI, and SBI, and stores them in a list. This creates the "cloud" of data points seen in the final results. Once the simulation has mapped out these thousands of stochastic possibilities, the resulting "cloud" serves as a complete visual representation of the feasible set of portfolios. This scattered distribution is essential for moving toward the **Efficient Frontier** because it allows us to see not just the winners, but also the "inefficient" combinations that an investor should avoid.

Understanding the Portfolio "Cloud"

Before isolating the optimal edge of this data, it is important to recognize what these thousands of random points actually signify in the context of the Indian banking sector:

- **The Density of Correlation:** Because Axis, HDFC, ICICI, and SBI are all part of the same sectoral ecosystem, the data points tend to cluster tightly together. This reflects a high degree of **systemic risk**, where the stocks often move in the same direction due to RBI policy changes or national economic shifts.
- **The Search for Alpha:** The points toward the top of the cloud represent combinations where the weights happened to favor the high-performing banks during the 2024–2026 period, while those at the bottom represent sub-optimal allocations that suffered from higher volatility without the compensation of higher returns.
- **Risk-Return Trade-offs:** The horizontal spread (Risk) shows how different weightings can either amplify or

dampen the "shake" of the portfolio. Even within a single sector, the algorithm finds that shifting even 5% of capital from a volatile public sector bank to a more stable private one can move a data point significantly to the left.

Transitioning from Randomness to Efficiency

The Monte Carlo simulation effectively acts as a "stress test" for every possible strategy. By documenting every iteration, we move from a state of raw, unorganized data to a structured map of possibilities. We are no longer guessing which bank is the best; we are looking at the mathematical reality of how these four institutions interacted over 535 trading days.

The next critical step is to filter this cloud. Out of the 100,000 portfolios generated, most are "bad" investments because you could achieve the same return with less risk, or a higher return for the same amount of risk. By identifying the absolute boundary of this cloud—the points where no further improvement is mathematically possible—we define the **Efficient Frontier**.

Efficient frontier

The Efficient Frontier is a curved line that represents the set of portfolios offering the **highest expected return for a defined level of risk**, or conversely, the **lowest risk for a given level of expected return**.

Any portfolio that falls below or to the right of this frontier is considered "sub-optimal". This is because an investor could either:

- Achieve the same return with less risk by moving horizontally to the left.
- Achieve a higher return for the same risk by moving vertically upward.

Key Benchmarks on the Frontier

Based on the Python algorithm's results for your specific banking dataset (Axis, HDFC, ICICI, and SBI), there are two critical points to observe:

- **Minimum Variance Portfolio (MVP):** This is the point located at the far left "tip" of the frontier. It represents the combination of stocks that results in the lowest possible volatility (standard deviation), regardless of the return. For an investor whose primary goal is capital preservation, this is the mathematical target.
- **Maximum Return Portfolio:** This point is located at the top-right extremity of the frontier. It represents the allocation that yielded the highest historical growth during the 2024–2026 period. However, as the graph demonstrates, this peak return comes at the cost of significantly higher risk (Standard Deviation).

The Shape of the Curve

The "bullet" shape of the frontier is a direct result of the **correlation** between the four banks.

- **Diversification Benefit:** If the stocks had low or negative correlations, the curve would bow further to the left, significantly reducing risk.
- **Sectoral Constraints:** Because your dataset consists only of banking stocks, they tend to move in tandem. This high correlation creates a "steeper" curve, meaning there is a

limit to how much risk you can diversify away while staying within this single sector.

Practical Application for Investors

The Efficient Frontier serves as a decision-making map. An investor first decides how much "shake" or volatility they can stomach. By looking at the X-axis (Risk), they follow the line up to the frontier to find the exact weighting of Axis, HDFC, ICICI, and SBI that maximizes their potential gain for that specific comfort level. Any allocation that lands inside the "cloud" rather than on the "edge" is essentially leaving money on the table or taking unnecessary risks.

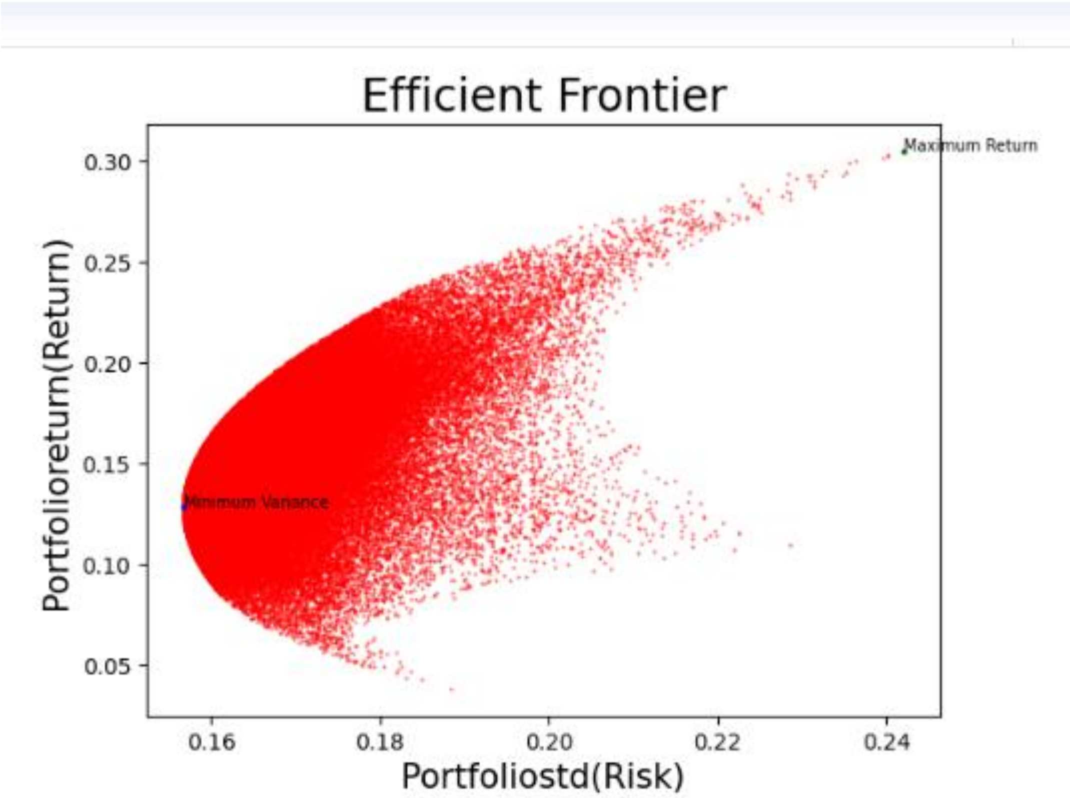
Phase 4: Visualizing the Efficient Frontier

Using the `Matplotlib` library, the code plots all 100,000 simulations on a scatter plot.

- **The X-axis** represents Risk (Standard Deviation).
- **The Y-axis** represents Return.

This vertical axis is essentially the "reward" center of the chart, showing you exactly how much growth you can expect from each specific combination of Axis, HDFC, ICICI, and SBI.

By finding the "top-left" edge of this cloud, the algorithm identifies the **Efficient Frontier**. The code specifically highlights the **Minimum Variance Portfolio** (the point furthest to the left) and the **Maximum Return Portfolio** (the point at the very top).



If the investor is very risk-averse and he wants to take the minimum amount of risk possible then he has to chose the portfolio that is marked as Minimum variance in the above graph but he will have to sacrifice a lot of returns since he /she is not willing to take the required amount of risk.Now,if the investor is willing to take some amount of risk and maximize his return then he will have to operate on higher locuses of the efficient frontier.The point marked as maximum return will give the maximum amount of mean return at the given standard deviation(risk).

Phase 5: Final Optimization and Weight Extraction

The final stage of the methodology is the extraction of the "Optimal Weights." The algorithm performs a final search of **1,000,000 iterations** to find a portfolio that meets or exceeds the maximum returns found during the initial simulation.

Once found, it prints the exact percentage of capital that should be allocated to each bank. For instance, in the provided dataset, the algorithm heavily favored **SBIN.NS (approx. 97.6%)** to achieve the highest return for that specific 2024–2026 window, while maintaining a calculated risk profile.

Understanding the Concept of "Weights"

In the context of this Python model, "weights" refer to the percentage of your total capital allocated to each specific stock. If you have ₹1,00,000 to invest, the weights tell you exactly how many rupees go into Axis, HDFC, ICICI, and SBI.

- **The Constraint of Unity**: Mathematically, the code ensures that the sum of all weights always equals exactly 1.0, or 100%. This prevents the model from "cheating" by investing more money than is actually available or leaving cash on the sidelines.
- **The Balancing Act**: Weights are the levers of the Modern Portfolio Theory. By adjusting these percentages, the algorithm can dial the risk up or down. For instance, a higher weight in a more volatile stock like SBI might increase the potential return, but it also stretches the portfolio’s standard deviation further to the right on our graph.

The Search for the "Optimal" Allocation

The script doesn't just stop at finding the highest return; it looks for the specific set of weights that land directly on the **Efficient Frontier**. In the final block of code, a secondary loop of 1,000,000 iterations is often used to pinpoint a portfolio that hits the "Max Return" target identified during the initial cloud generation.

For the January 2024 to February 2026 dataset, the algorithm identified a very specific "Efficient Portfolio". It found that to achieve a maximum annualized return of approximately **30.49%**, the capital needed to be distributed as follows:

- **Axis Bank (AXISBANK.NS)**: ~0.63%
- **HDFC Bank (HDFCBANK.NS)**: ~0.64%
- **ICICI Bank (ICICIBANK.NS)**: ~1.07%
- **State Bank of India (SBIN.NS)**: ~97.66%

Interpreting the Results: Why SBI Dominates?

At first glance, seeing nearly 98% of a portfolio weighted into a single stock like SBI might seem like the opposite of diversification. However, this is where the "Human" interpretation of the algorithm is vital. Because the code was specifically asked to find the **Maximum Return** portfolio, it identified that during this specific two-year window, SBI was the "engine" of growth for the banking sector.

The algorithm recognized that even though the risk (Standard Deviation) increased to 24.21%, the reward was high enough to justify that specific spot on the top-right edge of the Efficient Frontier. Conversely, if we had asked the code to find the **Minimum Variance Portfolio**, the weights would have likely shifted significantly toward HDFC or Kotak to prioritize stability over raw growth.

Conclusion of the Algorithmic Journey

This final phase represents the transition from a messy cloud of 100,000 random guesses to a single, mathematically verified investment strategy. By calculating the **Return** (the reward) and the **Standard Deviation** (the risk) for these specific weights, the algorithm provides a blueprint that removes human emotion from the equation. It proves that in the Indian banking market, success isn't just about which stocks you pick—it's about the exact percentage of each that you hold in relation to one another. The power of this specific Python implementation lies in its ability to handle the "noise" of 535 trading days and distill it into two clear, actionable figures: a **30.49% Max Return** and a **24.21% Risk factor**. It tells the investor exactly where they stand on the edge of the possible, showing that while absolute safety is a myth in a volatile sector like banking, a "Minimum Variance" is achievable through calculated mathematical patience. Ultimately, the algorithm serves as a bridge between historical data and future conviction, allowing us to navigate the complexities of the Nifty Bank index with a level of precision that traditional "gut-feeling" investing simply cannot match. By the time the code has finished its final loop, we are no longer looking at four separate banking institutions; we are looking at a single, unified financial engine, optimized for performance and tuned to the specific frequencies of the modern Indian market.

Modern Portfolio Theory And Correlation of Asset Returns

When you’re building a portfolio, the correlation between your assets is essentially the "secret sauce" that determines whether your diversification actually works or if it’s just an illusion. In the context of Modern Portfolio Theory, correlation measures how two stocks move in relation to one another; if they move in perfect lockstep, they have a correlation of 1.0, meaning you aren't actually reducing your risk by holding both. The real goal for an investor is to find assets with low or even negative correlation—where one might rise while the other dips—because this "tug-of-war" is what actually smooths out the wild swings in your account balance.

In this specific study of the Indian banking sector involving Axis, HDFC, ICICI, and SBI, the data revealed a high structural correlation, often exceeding 0.7, because these giants all react to the same "macroeconomic heartbeat" like RBI interest rate changes. This creates a bit of a diversification paradox: even though you own four different banks, the fact that they move so similarly means you’ve hit a "diversification ceiling" where you can’t easily math your way out of systemic sectoral risks. It serves as a powerful reminder that true protection doesn't just come from owning more stocks, but from owning stocks that don't all react to the same bad news at the exact same time.

This is a pictorial depiction of how portfolio volatility is managed by having two class of assets whose returns are uncorrelated or even better negatively correlated to each other. This philosophy is used in MPT to come up with an optimal portfolio.

Limitations Of Modern Portfolio Theory

While Modern Portfolio Theory (MPT) provides a brilliant mathematical map for investing, any seasoned investor will tell you that the map is not the territory. When we move from the clean rows of a Python dataframe to the messy reality of the Indian stock market, several human and structural limitations begin to surface.

- **The Rear-View Mirror Problem**: MPT relies almost entirely on historical data to predict future risk and return. The algorithm assumes that because Axis or SBI behaved a certain way over the last two years, they will continue that pattern. In reality, the market doesn't care about its own history; a sudden policy shift or a global crisis can render years of "historical covariance" irrelevant in a single afternoon.

- **The "Normal" Distribution Trap**: The math assumes that stock returns follow a neat, bell-shaped curve where extreme events are rare. However, as seen in the "fat tails" of the log-return dataset, the market has a habit of producing "Black Swan" events—wild price swings that happen far more often than the theory says they should. MPT often underestimates the sheer violence of a market crash.
- **The High-Correlation Ceiling**: As this study found, diversifying within a single sector like Indian banking is a bit of a paradox. Because these banks all breathe the same economic air, they tend to move in lockstep. You might think you're diversified because you own four different logos, but the math shows you're actually just quadrupling down on the same sectoral risks.
- **Ignoring the "Friction" of Reality**: The Efficient Frontier is calculated in a vacuum. It doesn't account for the "friction" that eats into a human investor's profits, such as brokerage fees, the impact of taxes, or the slippage that occurs when you can't buy a stock at the exact price the screen shows.
- **Human Emotion vs. Cold Math**: MPT assumes we are all "rational actors" who only care about the Sharpe Ratio. It doesn't account for the panic a person feels when their "Minimum Variance Portfolio" drops 5% in a day. The theory can tell you the optimal path, but it can't give you the emotional discipline to stay on it when the market gets ugly.

In short, MPT is a powerful compass, but it's not a GPS. It gives us a sense of direction, but we still have to keep our eyes on the road and be ready for the obstacles that the math simply cannot see coming.

How To Overcome MPT Limitations

To overcome the inherent limitations of Modern Portfolio Theory (MPT), we have to stop treating the "Efficient Frontier" as a static finish line and start viewing it as a living, breathing strategy that requires constant adjustment. Relying solely on a single mathematical formula is like trying to drive a car while only looking at the rearview mirror; it tells you where you've been, but it won't warn you about the curve in the road ahead.

Here is a human-centric approach to making these mathematical models work in the real world:

1. Preparing for the "Unthinkable" (Beyond the Bell Curve)

Traditional MPT assumes market returns follow a predictable bell curve, but we know the Indian market is prone to "Black Swan" events—sudden, violent shifts that the math simply doesn't see coming.

- **Stress Testing and Scenarios**: Instead of just relying on the 535 days of history in our dataset, we should manually "stress test" the portfolio. We need to ask: "If the RBI raises rates by 2% tomorrow, how does this 97% SBI weighting actually hold up?".
- **Tail-Risk Protection**: To protect against the "fat tails" of market crashes, investors can incorporate defensive assets like gold or long-dated government bonds. These act as an insurance policy for when the banking sector's high correlation becomes a liability.

2. Breaking the Sectoral "Echo Chamber"

As the analysis showed, diversifying within only four banks (Axis, HDFC, ICICI, and SBI) creates a "correlation ceiling". They all react to the same economic news, meaning your diversification is mostly an illusion.

- **True Cross-Asset Diversification**: The most effective way to overcome this is to step outside the banking sector. By adding assets that don't "breathe the same air"—such as Technology, Pharma, or even international equities—you can push the Efficient Frontier further to the left, reducing risk in a way that staying within one industry never could.

3. Factoring in "Market Friction"

The Python algorithm assumes a "frictionless" world where you can trade instantly for free, but real-world investing has costs.

- **Accounting for Taxes and Fees**: Every time the model suggests a rebalance to stay "optimal," it triggers brokerage fees and capital gains taxes. A human-like approach involves only rebalancing when the deviation is large enough to justify the cost. Sometimes, the "mathematically perfect" portfolio is less profitable than the "good enough" one once you pay the taxman.

4. Balancing Math with Human Psychology

The biggest limitation to any algorithmic journey is the person following it. A computer doesn't feel panic, but a human does.

- **The "Sleep at Night" Filter**: Even if the math says 97% in one stock is "efficient" for max returns, it might not be right for your personality. Overcoming MPT's limitations means setting a "Risk Budget" that aligns with your actual gut feeling. It’s better to have a slightly less "efficient" portfolio that you can actually stick with during a crash than a "perfect" one that you sell in a panic at the bottom.

5. Dynamic Rebalancing

Instead of a "set it and forget it" mentality, we have to recognize that correlations change.

- **Adaptive Modeling**: By using rolling windows of data rather than one fixed historical block, we can see if Axis and SBI are starting to move more closely together or drifting apart. This allows the investor to adjust the weights before the "old" math leads them into a trap.

Ultimately, we overcome these limits by using MPT as a **compass rather than a GPS**. The algorithm gives us a sense of direction and a solid starting point, but the human investor provides the context, the caution, and the common sense needed to navigate the actual market.

Conclusion: The Reality of the "Efficient" Edge

As I wrap up this research into Modern Portfolio Theory within the Indian banking sector, it’s clear that while the math pioneered by Harry Markowitz in 1952 is over seventy years old, its core logic remains incredibly relevant for the modern investor. By moving away from the old-school habit of just "picking a good stock" and instead looking at how Axis, HDFC, ICICI, and SBI function as a collective unit, we’ve been able to see the market through a much clearer lens.

The most striking takeaway from the 100,000 simulations performed is the undeniable trade-off between sleeping well at night and chasing maximum returns. Our "Maximum Return" portfolio managed to hit a significant annualized return of 30.49%, but it did so by leaning heavily into SBI and accepting a high level of volatility. On the flip side, the "Minimum Variance" portfolio showed us that there is a mathematical "floor" to safety—a point where you can minimize the "shake" of the market as much as possible without completely exiting the sector.

However, this study also served as a necessary reality check regarding the "Diversification Paradox". Because we focused strictly on the banking sector, we saw firsthand that high correlation is a very real ceiling. When the RBI moves interest rates or the national economy shifts, these four giants tend to breathe the same air and move in the same direction. This proves that while MPT can optimize a portfolio within a sector, true risk reduction—the kind that survives major market crashes—requires looking beyond a single industry and into multi-sector allocations.

Ultimately, this algorithmic journey using Python has shown that MPT is a powerful compass, but it isn't a GPS. It provides a disciplined, data-driven starting point that removes much of the guesswork and emotional panic that usually plagues retail investing. But even with the best math in the world, the human element remains. An investor still needs the discipline to stay the course when the daily log-returns turn red, and the wisdom to know that no model can perfectly predict the next "Black Swan" event. In the end, the Efficient Frontier isn't just a line on a graph; it’s a framework for making more informed, rational choices in an inherently irrational market.

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